Reset Early, Reset Often, Eliminate Models

How aggressively resetting connection tableaux can saturate the SAT solver and find an unsatisfiability proof faster.

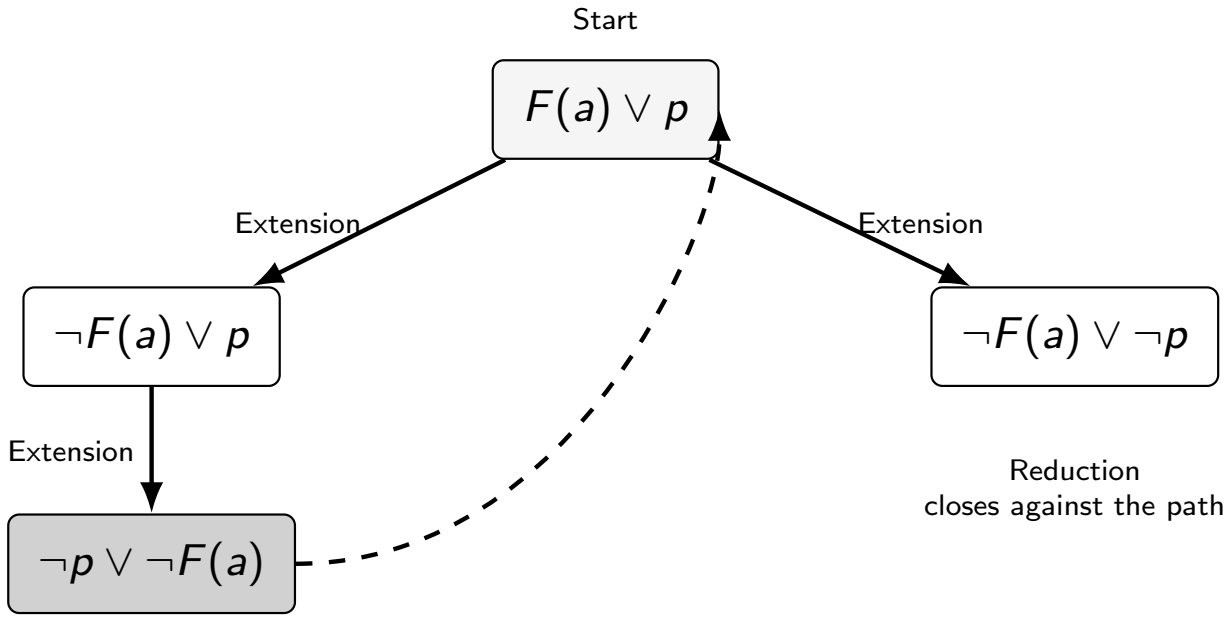
Martin S. Fixman[1], Fredrik Rømming[1], Sean B. Holden[1]

[1] UNIVERSITY OF CAMBRIDGE

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LeanCoP-style provers

Build one connection tableau by goal-directed search.



Proof-building rules
Start, Extension, and Reduction grow and close the tableau.

Search control
Backtracking revisits choices when a locally sound connection leads to a dead end.

Failed tableaux leave little information to reuse.

(Letz and Stenz 2001; Otten and Bibel 2003)

Keep a persistent propositional view of tableau search.

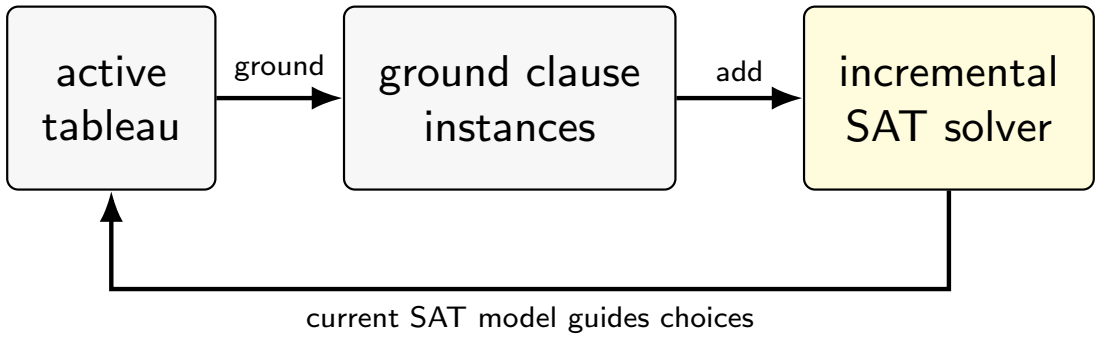


Tableau route

Close every open branch.

or

SAT route

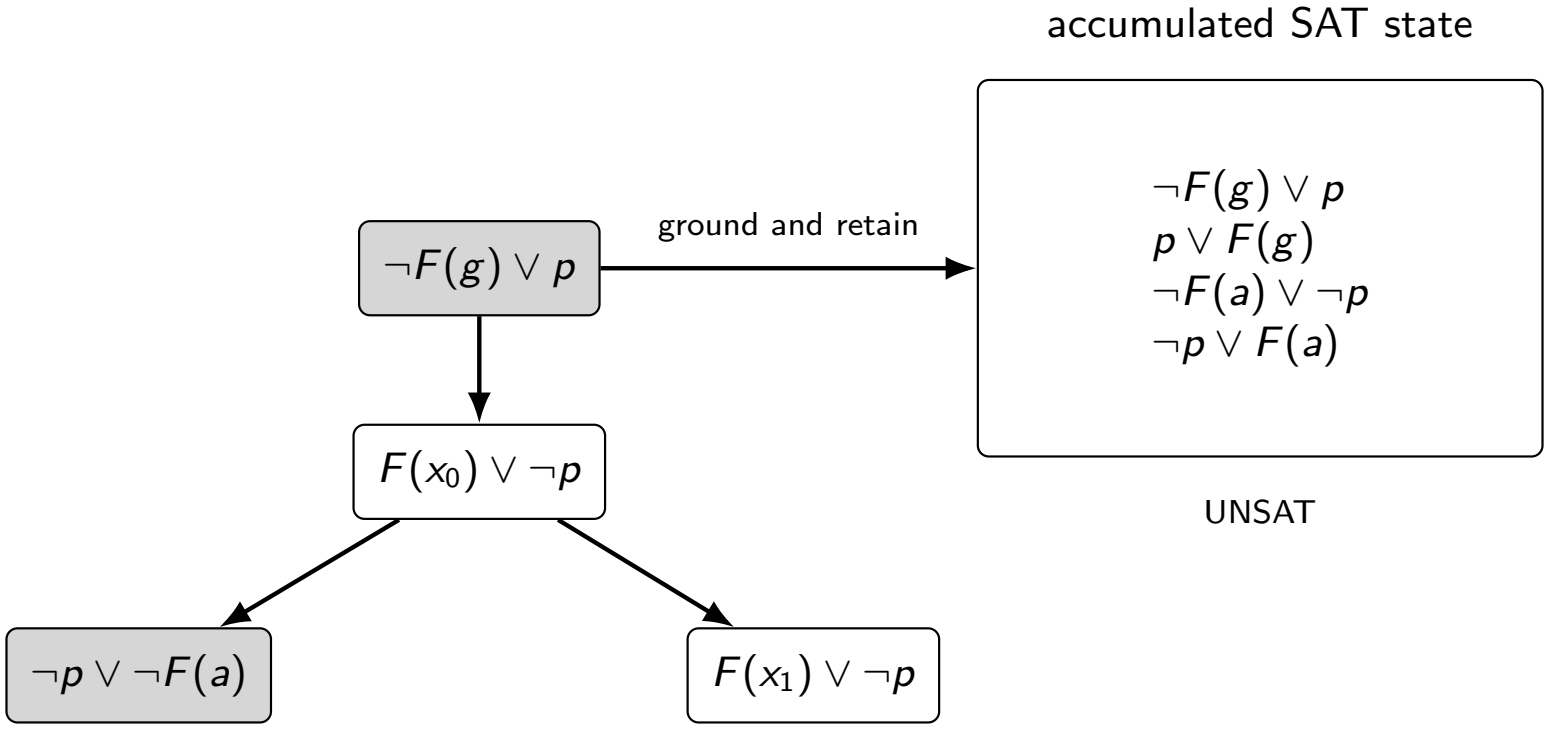
Accumulated ground clauses become UNSAT.

The SAT solver can finish the proof before the active tableau closes.

(Rawson and Reger 2021)

Why is SATCoP so good?

SYN052-1.p: the useful result of a failed branch is no longer lost.



leanCoP: **43 steps**
Must eventually close one tableau.

SATCoP: **17 steps**
Ground clauses from several attempts form an UNSAT core.

Failed search becomes useful work.

(Rawson and Reger 2021)

Improvements to SATCoP

SATCoP makes the SAT solver part of search control.

	SAT refutation	SAT-model guidance	Local backtracking
leanCoP			✓
SATCoP	✓	✓	✓

Explore more broadly

Randomise starts, extension choices, and clause-literal order.

Follow the SAT model

Prefer extensions whose selected literal is true in the current model.

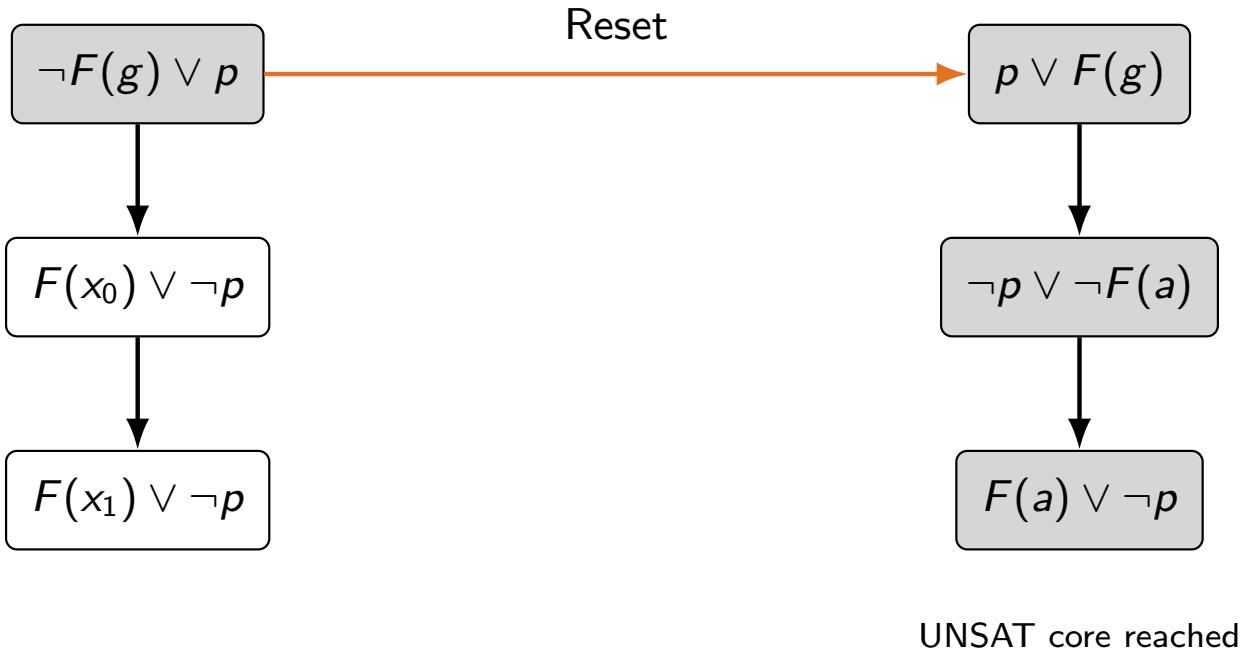
The SAT solver needs diverse ground clauses; local backtracking keeps exploring near the same tableau choices.



(Rawson and Reger 2021; Otten 2010; Rømming et al. 2023)

What if we stop doing regular tableau guidance?

When an attempt stops producing new ground clauses, use Reset instead of Backtrack.



SATCoP: **17 steps**
Run Backtrack through the current tableau.

SATResetCoP: **6 steps**
Run Reset to expose the UNSAT core faster.

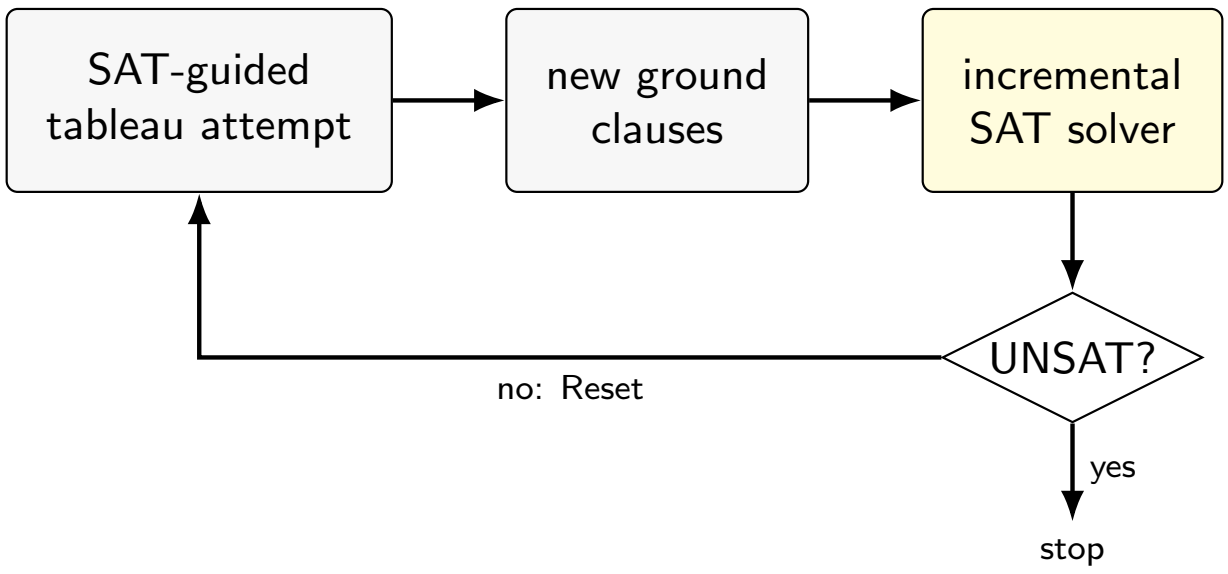
Use tableaux as **ground-clause generators** for SAT.

(Rawson and Reger 2021)

SATResetCoP

Guide tableau search to saturate the incremental SAT solver.

(Rømming et al. 2023; Biere et al. 2024)



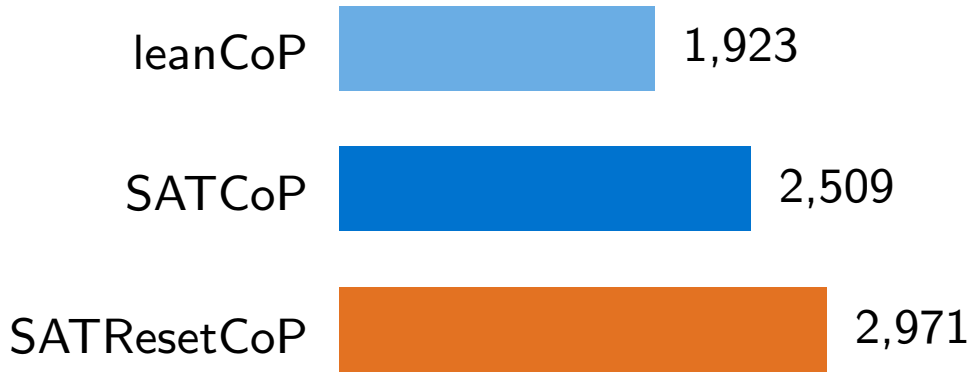
Choose
Randomise starts and extensions; prefer literals supported by the SAT model.

Accumulate
After Extension and Reduction, collect genuinely new ground clauses.

Reset
At a dead end, flush new clauses and run Reset; increase depth only after no progress.

General Results

TPTP problems solved



MPTP2078

616 problems

+12% over SATCoP

+49% over leanCoP

Aggressive Resets solve more problems, usually in fewer steps.

120 seconds and 1 million tableau steps per problem. The Reset policy also loses 136 TPTP problems solved by SATCoP: a schedule should keep both search modes.

(Sutcliffe 2017; Alama et al. 2014)

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